Uncertainty and Sentiment in ARK Company Filings

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Repository: github.com/victor4628/FRE7871-NLP

I examine financial negative language and uncertainty in 1,624 eligible 10-K/10-Q filings from 91 ARK-held issuers, filed during 2021-2025. The clearest evidence is a rise in negative language within annual-report issuers. Uncertainty trends depend on report type and weighting. Its association with subsequent volatility weakens after controlling for existing volatility. Four-day return estimates are imprecise; they do not establish either predictability or the absence of an effect.

Table 1 Sample construction

Security and issuer screen	Removed	Remain
Unique cleaned securities in frozen snapshot	0	124
SEC ticker-to-CIK mapping found	7	117
At least one 2021-2025 10-K/10-Q	24	93
Distinct issuers after multi-class consolidation	1	92

Unique filing screen	Removed	Remain
Downloaded security-filing records	0	1,702
Duplicate Alphabet records (retain GOOGL)	20	1,682
Valid acceptance timestamp	0	1682
At least 2,000 narrative words	0	1682
Exclude explicitly identified shell companies	58	1624

Market screen (separate branches)	Vol: removed / left	Return: removed / left
Starting unique filings	0 / 1624	0 / 1624
Complete 63-day pre-filing returns	32 / 1592	32 / 1592
Dated instantaneous common shares and positive market value	0 / 1592	0 / 1592
Complete positive pre-filing turnover	0 / 1592	0 / 1592
Complete outcome window and benchmark	1 / 1591	0 / 1592

The initial 31 losses are 7 unresolved SEC ticker mappings and 24 mapped securities without an eligible 10-K/10-Q in the window. Filing counts for those excluded securities are not observed and are not invented. The 93 eligible securities represent 92 issuers because GOOG and GOOGL share filings. Explicitly reported shells are excluded. The optional $3 price cutoff was removed at the student's request; low-priced stocks remain eligible. Legacy shell checkboxes and ordinary-share cover wording were corrected. All download and parsing failures were zero.

Measures and identification

The instructor-specified LM lists contain 2,355 negative words and 297 uncertainty words. They overlap on 40 words, so the concepts are distinct but their empirical measures are not independent. Ten nonzero Negative flags actually mark removed entries; the required list is retained for comparability, with an active-only list as a sensitivity.

For term i in document j, I implement equation (1) as:
$\mathbf{w_{ij} = [(1 + \ln tf_{ij}) / (1 + \ln a_j)] \ln(N / df_i)}$ for tf > 0; otherwise zero.
Here a_j is total tokens divided by distinct tokens, not total document length. A category score sums its term weights. Proportions divide category occurrences by all tokens. IDF is recomputed on the exact document corpus used by each regression, including restricted sensitivities. This is full-corpus, retrospective scoring, not an out-of-sample forecast.

Parsing retains visible inline-XBRL text and removes hidden resources and mostly numeric tables (digit share above 15%). The corrected parser retains 62,590,430 tokens across unique downloads versus 48,537,683 under the starter parser, which also deleted visible tagged narrative. The starter uppercase tokenizer is retained; there is no stemming or stopword removal.

Table 2 Summary statistics by form

Form / measure	N	Mean	SD	P25	Median	P75
10-K / Negative %	396	2.347	0.406	2.084	2.361	2.620
10-K / Negative TF-IDF	396	163.684	54.480	122.192	155.045	204.915
10-K / Uncertainty %	396	1.947	0.240	1.803	1.969	2.119
10-K / Uncertainty TF-IDF	396	26.643	8.120	21.062	25.542	31.201
10-Q / Negative %	1,228	2.133	0.948	1.285	1.952	3.020
10-Q / Negative TF-IDF	1,228	80.297	66.244	24.021	56.244	131.109
10-Q / Uncertainty %	1,228	1.829	0.587	1.327	1.669	2.419
10-Q / Uncertainty TF-IDF	1,228	13.505	7.271	7.788	11.909	18.719

Negative-uncertainty correlations: 10-K prop: 0.722; 10-K tfidf: 0.838; 10-Q prop: 0.851; 10-Q tfidf: 0.921. Shared risk topics, overlapping vocabulary and document structure can make the scores move together. The separate outcome tests do not assume statistical independence.

Event timing and controls

Day 0 is the first exchange session closing strictly after the SEC acceptance timestamp, including early closes; 921 core filings move from their filing date. Four-day return is stock buy-and-hold return over [0,3] minus SPY buy-and-hold return. Volatility is daily-return SD times sqrt(252): [-63,-1] before and [4,66] after, excluding the four-day reaction. All 63 returns are required; missing prices are never filled.

Size uses dated common shares from the same filing and day -1 nominal price; subsequent split adjustments are reversed. Turnover uses split-consistent pre-filing volume and the filing share count. Multiple common classes are summed and valued at the selected class price, an approximation. Weighted-average EPS shares are excluded from the main analysis. Models control for log size, log turnover, prior return, form, two-digit SIC and calendar-quarter effects; controlled models also include log prior volatility. Standard errors cluster by issuer.

Table 3 Words driving each measure

Shares below divide each word count by all occurrences of words on its own list, not by all words in the filings. The top 30 words account for 46.4% of Negative occurrences and 90.4% of Uncertainty occurrences. Uncertainty is particularly concentrated, which makes common-word downweighting consequential.

Rank	Negative word	List share	Uncertainty word	List share
1	LOSS	5.12%	MAY	35.22%
2	ADVERSELY	4.13%	COULD	18.54%
3	LOSSES	3.11%	RISK	4.98%
4	CLAIMS	2.97%	RISKS	4.48%
5	ADVERSE	2.77%	BELIEVE	2.93%
6	AGAINST	2.38%	APPROXIMATELY	2.56%
7	UNABLE	1.96%	ASSUMPTIONS	1.92%
8	LITIGATION	1.95%	INTANGIBLE	1.64%
9	HARM	1.87%	POSSIBLE	1.26%
10	FAILURE	1.81%	UNCERTAINTIES	1.16%
11	FAIL	1.34%	FLUCTUATIONS	1.12%
12	IMPAIRMENT	1.24%	MIGHT	1.06%
13	NEGATIVELY	1.14%	UNCERTAIN	1.02%
14	PENALTIES	1.14%	ANTICIPATED	0.99%
15	DIFFICULT	1.14%	PREDICT	0.95%
16	NEGATIVE	1.02%	VOLATILITY	0.94%
17	CRITICAL	1.01%	DEPEND	0.93%
18	DELAYS	0.94%	UNCERTAINTY	0.89%
19	DECLINE	0.92%	DIFFER	0.85%
20	DAMAGES	0.91%	ANTICIPATE	0.80%
21	RESTATED	0.85%	CONTINGENT	0.77%
22	LIMITATIONS	0.85%	EXPOSURE	0.77%
23	DELAY	0.84%	VARIABLE	0.69%
24	VOLATILITY	0.77%	DEPENDS	0.68%
25	CHALLENGES	0.76%	PENDING	0.65%
26	FINES	0.72%	DEPENDENT	0.58%
27	CLOSING	0.69%	CONTINGENCIES	0.55%
28	HARMED	0.69%	PROBABLE	0.51%
29	DISRUPTIONS	0.69%	ASSUMED	0.51%
30	INVESTIGATIONS	0.66%	VARY	0.48%

A frequently used word can dominate proportional counts yet convey little cross-document distinction. Equation (1) reduces that contribution through document frequency, while log term frequency dampens repetition. A high TF-IDF value can also reflect unusual vocabulary and document structure; it is not a probability of bad news.

Figure 1 Tone and market uncertainty

Each line first averages filings within issuer, form and filing quarter, then weights issuers equally. Annual and quarterly reports remain separate. VIX is the quarterly mean on the right axis; it is a market comparison, not the dependent variable in the firm-volatility regressions.

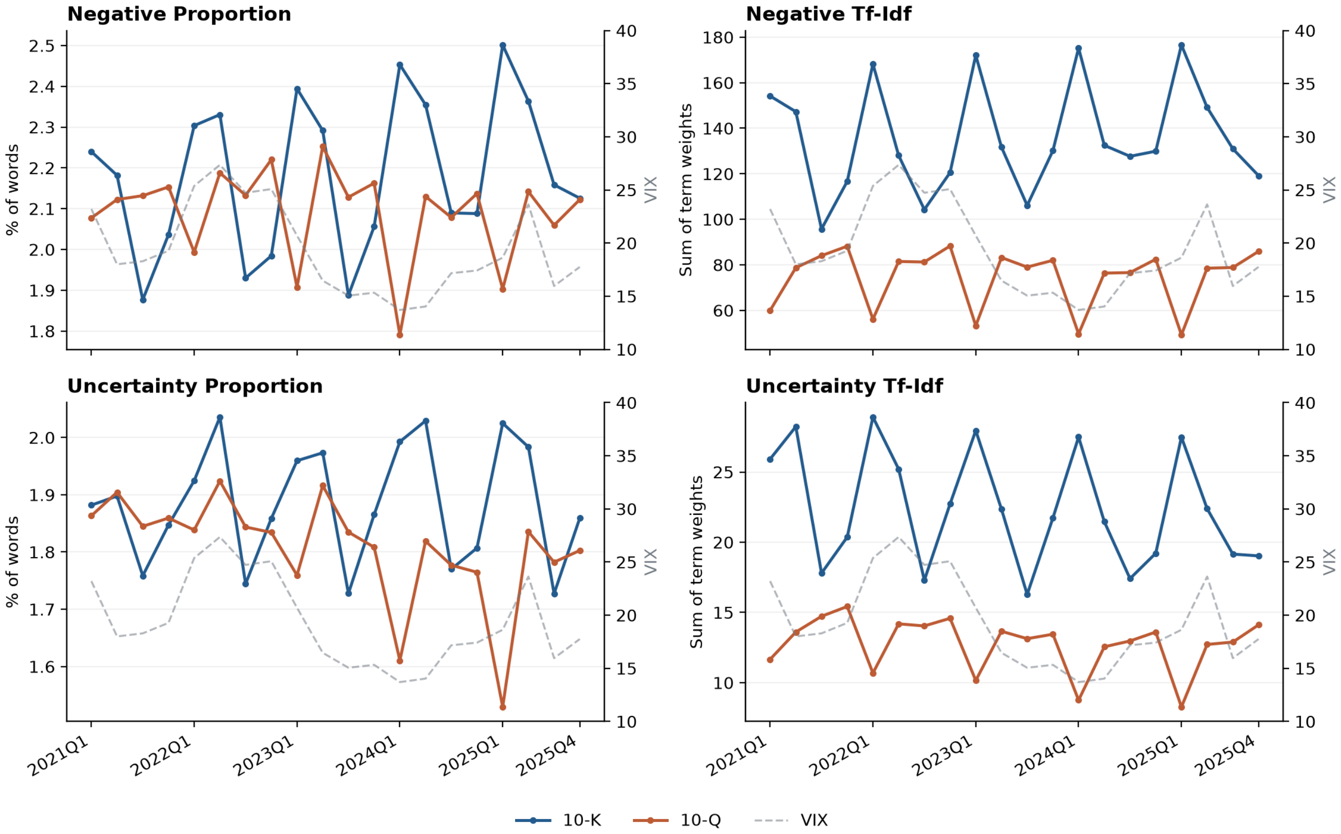

Table 4 Aggregate and within-issuer trends

Form	Measure	Agg. slope	OLS t	NW t	Within slope	p	Holm p
10-K	Negative %	0.0318	6.53	9.50	0.0345	<0.001	<0.001
10-K	Negative TF-IDF	0.0176	2.85	4.73	0.0133	0.002	0.007
10-K	Uncertainty %	0.0119	1.83	1.73	0.0249	<0.001	<0.001
10-K	Uncertainty TF-IDF	-0.0124	-1.44	-2.23	-0.0077	0.077	0.154
10-Q	Negative %	-0.0060	-2.57	-3.35	-0.0042	0.332	0.332
10-Q	Negative TF-IDF	-0.0062	-4.16	-4.06	-0.0072	0.113	0.154
10-Q	Uncertainty %	-0.0161	-4.30	-4.52	-0.0109	0.019	0.038
10-Q	Uncertainty TF-IDF	-0.0179	-6.28	-6.16	-0.0181	<0.001	0.001

Slopes are SDs per quarter. Aggregate models include quarter-of-year effects, with Bartlett Newey-West SEs (4 lags, finite-sample correction) alongside naive OLS t statistics. Within-issuer models include issuer and seasonal effects and issuer-clustered SEs; issuers observed in only one quarter are omitted from that test. Holm adjustment covers the four within-issuer tests separately for each weighting scheme. Redundant fixed-effect columns are removed algebraically.

The within-issuer results, rather than the pooled chart, support the main trend interpretation. Annual-report negative language rises under both weights. Uncertainty does not have one universal trend: annual-report proportional uncertainty rises while its TF-IDF counterpart does not; quarterly-report uncertainty declines. Changing report mix and issuer mix therefore cannot be ignored. With only 20 aggregate quarters, even corrected aggregate inference is fragile.

Table 5 Uncertainty and subsequent volatility

The dependent variable is log annualized post-filing volatility. Each coefficient is for a one-SD increase in the uncertainty score. Paired models share the same complete-case sample and IDF corpus; only the pre-volatility control changes.

Weight	Prior vol.	Coefficient	Cluster SE	p	Holm p	N
Proportion	No	0.0334	0.0168	0.050	NA	1,591
Proportion	Yes	0.0174	0.0086	0.045	0.136	1,591
TF-IDF	No	0.0604	0.0238	0.013	NA	1,591
TF-IDF	Yes	0.0264	0.0120	0.030	0.119	1,591

Proportional uncertainty falls from 0.0334 to 0.0174 after controlling for prior volatility, a 48.0% attenuation. The controlled estimate corresponds to about 1.8% higher volatility per SD. TF-IDF uncertainty falls from 0.0604 to 0.0264 after controlling for prior volatility, a 56.3% attenuation. The controlled estimate corresponds to about 2.7% higher volatility per SD.

The attenuation is the central result: hedged language partly describes companies that were already volatile. The controlled coefficients are conditional associations, not evidence that the words cause volatility. They are modest and their family-adjusted p values are less compelling than the unadjusted tests. Holm adjustment here covers the two controlled volatility and two return tests together.

Declared robustness checks

Specification	Prop. coefficient	p	TF-IDF coefficient	p
10-K	0.0498	0.015	0.0325	0.108
10-Q	0.0200	0.035	0.0174	0.100
firm FE	-0.0092	0.367	-0.0057	0.622
two way	0.0174	0.035	0.0264	0.035

The form-specific rows re-estimate IDF within that form. Firm effects replace industry effects. Two-way clustering uses issuer and calendar quarter; there are only 20 quarter clusters. The two-way covariance can have negative nuisance-parameter variances in this finite sample; these are logged, not set to zero, and target coefficients with invalid variances are reported as NA. The complete coefficient tables preserve all estimated specifications.

Replacing industry effects with issuer effects makes both uncertainty coefficients negative and insignificant. The positive pooled association therefore depends on between-issuer differences; robust incremental prediction within issuers is not established. Trend, volatility and return tests answer different questions. Sample selection and overlapping outcome windows further limit interpretation.

Table 6 Sentiment and four-day excess returns

Before interpreting significance, consider precision. Approximate 80%-power minimum detectable effects are 0.92 and 1.30 percentage points per SD for proportional and TF-IDF sentiment, using (cluster-t critical + 0.842) times the clustered SE. This is a design-precision diagnostic, not observed power.

Regressor / statistic	Proportion	TF-IDF
Negative tone (1 SD)	-0.460 (0.324)	-0.670 (0.459)
Log market value	-0.434 (0.310)	-0.467 (0.315)
Log turnover	0.659 (0.539)	0.643 (0.543)
Prior 63-day return	-0.483 (1.586)	-0.469 (1.582)
Log prior volatility	-1.950 (1.202)	-1.915 (1.219)
N	1,592	1,592
Issuer clusters	90	90
Adjusted R squared	0.014	0.014
Tone p	0.160	0.148
Tone Holm p	0.295	0.295

Dependent variable: stock minus SPY buy-and-hold return, in percentage points. Parentheses contain issuer-clustered SEs. Form, two-digit SIC and calendar-quarter effects are included but not printed. Both tone coefficients are negative and statistically imprecise. A null here is compatible with economically nontrivial effects and does not invalidate the separate trend or volatility tests.

Return sensitivities (coefficient / p)

Specification	Proportion	TF-IDF
10-K	-0.228 / 0.728	-0.260 / 0.719
10-Q	-0.588 / 0.138	-0.717 / 0.142
firm FE	-1.024 / 0.111	-1.139 / 0.164
two way	-0.460 / 0.160	-0.670 / 0.223
ARKK	-0.393 / 0.229	-0.512 / 0.276
active negative	-0.446 / 0.169	-0.659 / 0.153

Limits and next step

The 124-security universe is selected from 2026 holdings snapshots, not historical ARK ownership. Only 78 of 91 retained issuers appear in all five filing years. The number of formerly held firms missing from this snapshot cannot be established from the supplied data. Survivorship can distort trends as well as return tests; fixed effects do not remove this selection. Other limits are 20 aggregate quarters, full-corpus rather than live scoring, noisy daily event timing, multi-class valuation proxies, and overlapping post-filing windows. My highest-priority extension is a historical holdings universe with inclusion dates, followed by a genuinely held-out forecasting test.

Sources: Loughran and McDonald (2011), Journal of Finance 66(1), 35-65, equation (1); Fall 2026 assignment brief; instructor starter repository; LM dictionary documentation; Yahoo price-adjustment documentation; exchange_calendars; statsmodels. AI assistance is disclosed in AI_USE.md.